🧭 M003 Guide — Counting (x, y, z) · Advanced

Topic: Before 3D · **Course:** 3D Coordinates · **Level:** Advanced · **Time:** about 30 minutes

Pixel art used **two** numbers. A 3D build needs **three**. They always come in the same order:

1st = X	2nd = Y	3rd = Z
left → right	down → up	back → forward
starts at 0	starts at 0	starts at 0

The three rulers meet on **one block** — your home spot. That block is **(0, 0, 0)**. Every number is counted **from there**, so a coordinate is really three instructions: how far right, how far up, how far forward.

> 🔴 Stand on your **home spot** (red block) every run. Move your feet, move your build.

1 · Build Your Grid 🎨

Make a striped ruler for each axis, so you can **see** the numbers.

One `fill` makes a solid box between **two corners**. The floor is a flat box one block **below** your feet: from **(0, -1, 0)** out to **(15, -1, 15)**.

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(0, -1, 0),
        pos(15, -1, 15),
        FillOperation.REPLACE)
player.on_chat("grid", on_run)
```

Write the two `fill` lines that draw a yellow line 5 to the right (x = 5) and a yellow line 5 forward (z = 5), both across the whole floor.

The floor sits at y = -1, not y = 0. Explain what y = 0 is, and what would happen if you filled the floor there instead. (2–3 sentences.)

Now place the three rulers by hand — **yellow, black, yellow, black** all the way. All three start on the corner block: **x** to your right, **z** forward, **y** straight up.

2 · Predict 🔮

Read each box. Write what you will see, **where it sits, and the reason**.

```
fill RED from (0, 0, 0) to (5, 0, 0)
```

Only the 1st number changes. Which ruler is this line lying on, and which way does it run?

```
fill RED from (0, 0, 0) to (0, 5, 0)
fill RED from (0, 0, 0) to (0, 0, 5)
```

Two lines from the same corner. Describe the shape they make together, and name the corner's coordinate.

```
fill RED from (3, 0, 4) to (3, 0, 4)
```

Both corners are the same. How many blocks is this, and why?

```
place red block at (-1, 0, 0)
```

x counts to your right, so 1 is one step right. This one is negative. Where does the block land?

3 · Spot the Bug 🐛

Each one is broken. Read what it should do, write the fixed code, then explain.

Bug A — This should lay the white floor **under** your feet. It fills at **y = 0** instead, so the blocks land at feet height and swallow the home spot.

```
fill WHITE_CONCRETE from (0, 0, 0) to (15, 0, 15)
```

Hint: the **2nd number (y)** is height, and 0 is standing height. What is one block lower?

Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

Bug B — Your friend thinks the corner block is **(1, 1, 1)**, so they count every number from there. Their whole build ends up one step right, one block up, and one step forward from where it belongs — and it floats.

```
place red block at (5, 1, 5)   # they wanted the block 4 right, on the ground, 4 forward
```

Hint: the corner is where the three rulers **meet**.

Write the fixed code:

Which of the three numbers is the one lifting the build off the ground? Explain how you know. (2–3 sentences.)

Bug C *(the tricky one)* — This yellow block should sit further along the x ruler, at **x = 8**. The code uses **x = -8**, so it lands on the wrong side of the player entirely.

```
place yellow block at (-8, 3, 4)
```

Hint: which direction does a **negative** x count?

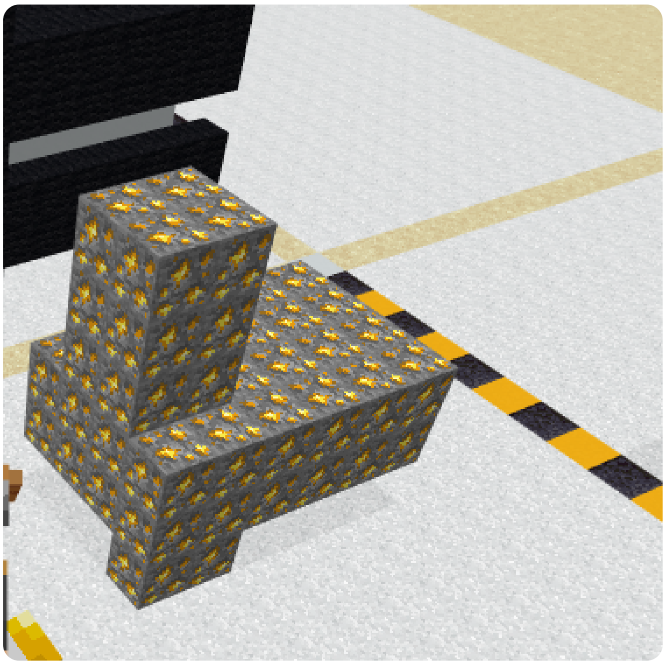
Write the fixed code:

Why was it wrong? Why does your fix work? (2–3 sentences.)

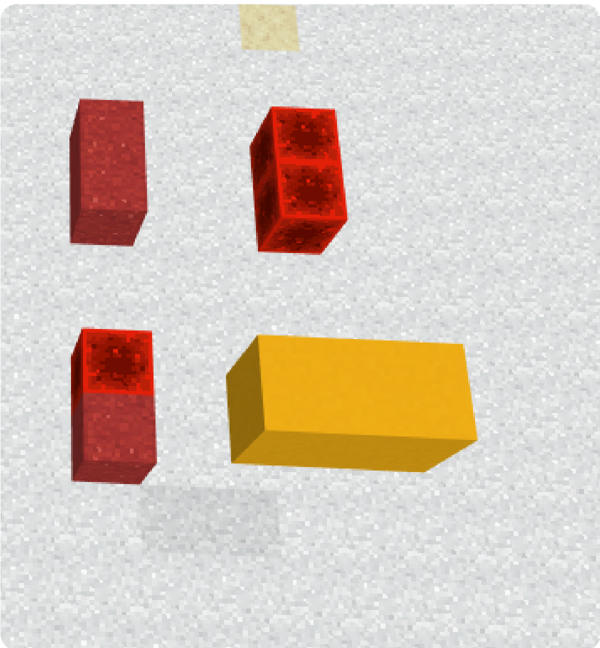
4 · Show Your Work 📸 🎥

Warm-up: stand on your home spot and place one block at **(3, 0, 3)**. Check it against all three rulers.

Open the world. Part of the shape is **already built** for you:



The rest of the blocks are **missing**. Here they are, spread apart so you can see each one:



Finish the shape. Read every missing block off the model and count from the corner — x, then y, then z.

Plan before you build. Write each missing block's coordinate. Where blocks sit in a straight line, use one `fill` between the two end corners instead of placing them one by one.

Write the code you ran.

> 🧱 The corner block is **(0, 0, 0)**. Use the same blocks you see in the world.
>
> ✨ **Modify challenge:** after it works, change one thing of your own — a new colour, an extra block, a mirrored copy further along z. Write what you changed.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Count one block out loud — x, then y, then z.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.